

Managed Leased Line Network

With the burgeoning need for the leased line, managed leased line networks (MLLNs) are being adopted and preferred by several businesses for fulfilling their telecommunication needs. Let us go through the technology of MLLNs, their advantages, usage, basic configuration, working, equipment involved, etc

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In the present age of globalisation, businesses are spread all around the world. In order to operate the business smoothly, there is a strong need for a dedicated telecom link that can connect geographically spread out locations of several businesses. The telecom facility may be either shared such as dial-up lines or non-shared such as leased lines and managed leased line networks (MLLNs). Obviously, businesses prefer a highly secured dedicated non-shared communication facility that is available all the time.

A leased line is a dedicated link or telecommunication path provided between two fixed locations, which is made available round the clock for use by the designated user (an individual or a company). Leased lines are provided to users for internal communication between their different business centres/offices/factories at various locations within a city or different cities on a point-to-point basis or on a network basis.

These leased lines can carry voice, data and video and can be used for connecting telephone sets, computers, electronic private automatic branch exchange (EPABX) and for establishing a virtual private network (VPN). These lines are exclusive for the designated user and are particularly not shared in common amongst multiple users as dial-up lines.

Such ordinary leased lines are not very efficient and have lots of disadvantages such as limited range of services, bandwidth support only up to 64kbps (2.4kbps, 4.8kbps, 9.6kbps and 64kbps), no support for n×64kbps

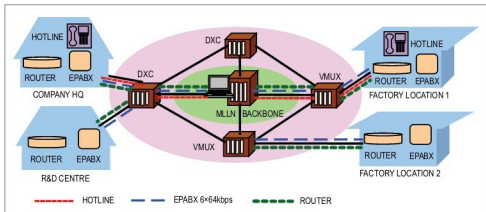


Fig. 1: A typical MLLN set-up

bandwidth provisioning, cumbersome operation and vendor-specific equipment.

Other disadvantages are first-generation network elements with minimal maintenance features, difficult fault isolation and rectification, no centralised alarm or performance statistics monitoring, no health diagnostics for the network, high lead time for new circuit installation, manual verification of resources needed, cross-connection of individual circuits done at the channel level, poor customer satisfaction, no proactive maintenance, high mean time to repair (MTTR), no guaranteed quality of service (QoS), no way to measure circuit quality or generate customised performance reports, poor network reliability and availability, non-redundant network elements, no alternate routing of circuits in case of failure and no centralised network management system (NMS).

Applications of MLLN

Managed leased line network (MLLN) is a system that can provide leased line connectivity, that is, a dedicated telecommunication path between two fixed points. It is an integrated, fully

managed, multi-service digital network platform through which a service provider can offer a wide range of services at an optimal cost to business subscribers. MLLN allows the service provider to keep an end-to-end control and monitor over the leased line and hence provide guarantees of uptime of the circuit.

With the use of managed leased line circuits by various sectors such as banking and financial institutions, stock markets, news and print media industry, broadcasting houses and Internet service providers (ISPs), people of all the sections are benefitted by way of accessibility of bank accounts from anywhere, instant news coverage, faster Internet access, etc.

The applications/services offered by MLLN are:

- 1. Speech circuits (hot line or P-wire).** Dedicated telecom links for speech, say, hot line for voices between two different locations is established by local or long-distance circuits within a city or between two different cities. The terminating equipment at both ends is telephone set without dialling facility. In such a connection, both-way signalling and speech is possible.